

Subject: Burgers Equation with Viscous and Artificial-Viscosity Shock

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Date : 2026-08-20 23:02:10

1 Problem

Consider the inviscid Burgers equation

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0, \quad 0 < x < L_x, \quad (1)$$

with boundary conditions

$$u(0, t) = a, \quad u(L_x, t) = b. \quad (2)$$

For the time-dependent problem, an initial condition

$$u(x, 0) = u_0(x) \quad (3)$$

is also required. The equation can be written in conservation form as

$$u_t + \left(\frac{u^2}{2} \right)_x = 0. \quad (4)$$

2 Characteristics

Along a characteristic,

$$\frac{dx}{dt} = u, \quad \frac{du}{dt} = 0. \quad (5)$$

Thus a constant value u_0 propagates with speed u_0 :

$$x = x_0 + u_0 t. \quad (6)$$

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3 Shock solution

For a shock connecting a left state a to a right state b , the Rankine–Hugoniot shock speed is

$$s = \frac{f(a) - f(b)}{a - b}, \quad f(u) = \frac{u^2}{2}. \quad (7)$$

Consequently,

$$\boxed{s = \frac{a + b}{2}}. \quad (8)$$

The shock position is therefore

$$x_s(t) = x_s(0) + \frac{a + b}{2} t. \quad (9)$$

A stationary shock requires

$$\boxed{a + b = 0} \quad (10)$$

or

$$\boxed{b = -a}. \quad (11)$$

For this case, the inviscid stationary solution is

$$u(x) = \begin{cases} a, & 0 \leq x < x_s, \\ b, & x_s < x \leq L_x. \end{cases} \quad (12)$$

The inviscid equation alone does not determine x_s when $b = -a$; any shock location inside the domain satisfies the stationary Rankine–Hugoniot condition.

For the frequently used case

$$a = 1, \quad b = -1, \quad (13)$$

the shock speed is

$$s = 0. \quad (14)$$

4 Viscous Burgers equation

5 Introduce a constant viscosity,

$$\boxed{u_t + uu_x = \nu_{\text{art}} u_{xx}.} \quad (15)$$

For a stationary shock,

$$uu_x = \nu_{\text{art}} u_{xx}. \quad (16)$$

Integrating once gives

$$\nu_{\text{art}} u_x = \frac{u^2}{2} + C. \quad (17)$$

For the symmetric case $a > 0, b = -a$,

$$C = -\frac{a^2}{2}, \quad (18)$$

so

$$\nu_{\text{art}} u_x = \frac{1}{2}(a^2 - u^2). \quad (19)$$

The viscous shock solution connecting a on the left to $-a$ on the right is

$$\boxed{u(x) = -a \tanh \left[\frac{a}{2\nu_{\text{art}}} (x - x_s) \right]}. \quad (20)$$

For $a = 1$,

$$u(x) = -\tanh\left[\frac{x - x_s}{2\nu_{\text{art}}}\right]. \quad (21)$$

The characteristic shock length scale is approximately

$$\ell_s \sim 2\nu_{\text{art}} \quad (22)$$

for $a = 1$, and more generally

$$\ell_s \sim \frac{2\nu_{\text{art}}}{a}. \quad (23)$$

5 General viscous travelling shock

For arbitrary left and right states $a > b$, the travelling viscous profile is approximately

$$u(x, t) = \frac{a+b}{2} - \frac{a-b}{2} \tanh\left[\frac{a-b}{4\nu_{\text{art}}}(x - x_s(t))\right], \quad (24)$$

where

$$x_s(t) = x_s(0) + \frac{a+b}{2}t. \quad (25)$$

For $a = 1$, $b = -1$, this reduces to

$$u(x) = -\tanh\left[\frac{x - x_s}{2\nu_{\text{art}}}\right]. \quad (26)$$

6 Artificial-viscosity discretisation

The artificial-viscosity matrix contribution used in the numerical scheme is

$$c_\psi \begin{bmatrix} -1 & 2 & -1 \end{bmatrix}. \quad (27)$$

Its action on the solution is

$$-c_\psi u_{i-1} + 2c_\psi u_i - c_\psi u_{i+1}. \quad (28)$$

Using the Taylor expansion

$$u_{i-1} - 2u_i + u_{i+1} = \Delta x^2 u_{xx} + \frac{\Delta x^4}{12} u_{xxxx} + O(\Delta x^6), \quad (29)$$

the artificial-viscosity contribution becomes

$$-c_\psi \Delta x^2 u_{xx} - \frac{c_\psi \Delta x^4}{12} u_{xxxx} + O(\Delta x^6). \quad (30)$$

If the matrix is normalized so that this term represents a standard diffusion term, the leading coefficient is associated with

$$\nu_{\text{art}} \sim c_\psi \Delta x^2. \quad (31)$$

However, the complete scheme contains an additional right-hand-side term and therefore this identification should not be made without considering the complete discrete equation.

7 Mass matrix used in the scheme

The mass stencil is

$$m_{\text{mass}} = \left[\frac{1}{8}, \frac{6}{8}, \frac{1}{8} \right]. \quad (32)$$

Thus the mass contribution is

$$\frac{1}{8}u_{i-1} + \frac{6}{8}u_i + \frac{1}{8}u_{i+1}. \quad (33)$$

Its Taylor expansion is

$$\frac{1}{8}u_{i-1} + \frac{6}{8}u_i + \frac{1}{8}u_{i+1} = u_i + \frac{1}{8}\Delta x^2 u_{xx} + \frac{1}{96}\Delta x^4 u_{xxxx} + O(\Delta x^6). \quad (34)$$

Combining this with the artificial-viscosity matrix gives

$$\begin{aligned} Au &= u + \left(\frac{1}{8} - c_\psi \right) \Delta x^2 u_{xx} \\ &\quad + \left(\frac{1}{96} - \frac{c_\psi}{12} \right) \Delta x^4 u_{xxxx} + O(\Delta x^6). \end{aligned} \quad (35)$$

Hence

$$Au = u + \left(\frac{1}{8} - c_\psi \right) \Delta x^2 u_{xx} + O(\Delta x^4). \quad (36)$$

This shows that the artificial-viscosity operator competes with the second-derivative contribution already present in the mass stencil.

8 Additional c_E contribution

The numerical scheme uses

$$c_E = c_\psi^2 \frac{\pi}{2}. \quad (37)$$

The right-hand side is

$$\text{rhs}_i = c_E \left(\frac{1}{2} \Delta x |u_{xx}| \right). \quad (38)$$

Consequently,

$$\text{rhs}_i = \frac{\pi}{4} c_\psi^2 \Delta x |u_{xx}|. \quad (39)$$

This means that the complete artificial-viscosity mechanism is not equivalent to a simple constant-viscosity Burgers equation. It is solution dependent because it contains $|u_{xx}|$.

An effective viscosity estimate based on interpreting this correction as a diffusive flux can be written schematically as

$$\nu_{\text{eff}}(x) \sim \frac{\pi}{4} c_\psi^2 \Delta x \frac{|u_{xx}|}{|u_x|}, \quad (40)$$

where this relation is an interpretation rather than an exact identification unless the complete time-stepping equation is specified.

9 Shock-width estimate

If one interprets the c_E contribution as producing an effective viscosity

$$\nu_{\text{eff}} \approx \frac{\pi}{4} c_\psi^2 \Delta x, \quad (41)$$

then for $a = 1$ the characteristic shock length is approximately

$$\delta \approx 2\nu_{\text{eff}} = \frac{\pi}{2} c_\psi^2 \Delta x. \quad (42)$$

In terms of grid cells,

$$N_s = \frac{\delta}{\Delta x}, \quad (43)$$

so

$$N_s \approx \frac{\pi}{2} c_\psi^2. \quad (44)$$

Solving for c_ψ gives

$$c_\psi \approx \sqrt{\frac{2N_s}{\pi}}. \quad (45)$$

Some example values are

Desired shock width	c_ψ
2 cells	1.13
3 cells	1.38
4 cells	1.60
5 cells	1.78
6 cells	1.95

Thus a reasonable starting value for a shock of about four grid cells is

$$c_\psi \approx 1.6. \quad (46)$$

These values are estimates because the exact effective viscosity depends on how the matrix A and the right-hand side are used in the time-stepping equation.

10 Original C++ artificial-viscosity code

The relevant code discussed was

```
double c_error = c_psi;
double c_E = c_psi * c_psi * std::numbers::pi / 2.0;

for (size_t i = 1; i < nx - 1; ++i)
{
    A.coeffRef(i, i - 1) = m_mass[0] - c_error;
    A.coeffRef(i, i) = m_mass[1] + 2. * c_error;
    A.coeffRef(i, i + 1) = m_mass[2] - c_error;

    rhs[i] = c_E * (0.5 * dx * std::abs(u_xixi[i]));
}
```

with

$$m_{\text{mass}} = \left[\frac{1}{8}, \frac{6}{8}, \frac{1}{8} \right]. \quad (47)$$

The exact effective viscosity cannot be determined uniquely from these lines alone. The remaining required information is the complete equation in which A and rhs are used, particularly the Eigen solve and the time-stepping terms.

11 Recommended interpretation

For a controlled numerical shock in Burgers' equation, a useful target is a shock spanning approximately 3–5 grid cells. The artificial viscosity should be strong near the shock and weak in smooth regions.

A common form is

$$u_t + uu_x = \frac{\partial}{\partial x} (\nu_{\text{art}} u_x), \quad (48)$$

with a solution-dependent viscosity such as

$$\nu_{\text{art}} = C_\nu \Delta x |u| \quad (49)$$

or a shock detector based on the local gradient or curvature. For the stationary shock

$$a = 1, \quad b = -1,$$

the ideal reference solution remains

$$u(x) = -\tanh \left[\frac{x - x_s}{2\nu_{\text{art}}} \right]. \quad (50)$$

The numerical solution can be compared against this profile to quantify shock width and wiggles.

12 Conclusion

For

$$u_t + uu_x = 0, \quad u(0, t) = a, \quad u(L_x, t) = b,$$

the inviscid shock speed is

$$s = \frac{a + b}{2}. \quad (51)$$

A stationary shock requires

$$b = -a. \quad (52)$$

With constant viscosity, the stationary viscous shock for $a > 0$ and $b = -a$ is

$$u(x) = -a \tanh \left[\frac{a(x - x_s)}{2\nu_{\text{art}}} \right]. \quad (53)$$

For the numerical scheme discussed here,

$$m_{\text{mass}} = \left[\frac{1}{8}, \frac{6}{8}, \frac{1}{8} \right],$$

and the artificial-viscosity stencil is

$$c_\psi[-1, 2, -1].$$

Its leading Taylor contribution is proportional to

$$-c_\psi \Delta x^2 u_{xx},$$

while the additional right-hand-side correction is

$$\frac{\pi}{4} c_\psi^2 \Delta x |u_{xx}|.$$

The exact continuous equation represented by the complete numerical scheme requires the remaining time-stepping/linear-system equations.